Smart Pesticide Residue Detection & Multi-Sensor Produce Quality Screening

Comprehensive Software Engineering, Experimental Results & Viva Exam Technical Report

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1. Project Overview & System Motivation

Chemical pesticide residues on post-harvest fruits and vegetables pose severe chronic health hazards, including endocrine disruption, neurotoxicity, and carcinogenesis. Conventional analytical quantification methods—principally Gas Chromatography-Mass Spectrometry (GC-MS) and Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS)—offer high precision but suffer from high capital expenditure, complex solvent extraction (e.g., QuEChERS protocols), multi-day turnaround times, and complete reliance on centralized laboratory infrastructure.

The **Smart Pesticide Residue Detection System** provides an edge-computed, point-of-consumption screening platform deployed directly on an ultra-low-power **STM32F407VET6** ARM Cortex-M4 microcontroller. The system fuses computer vision (OV2640 DVP camera running an INT8-quantized deep convolutional neural network) with multi-modal physical and chemical sensing (TCS34725 optical spectrophotometer, MQ135 electro-catalytic gas sensor, analog glass-electrode potentiometric pH probe, and DHT22 thermo-hygrometric sensor).

```
flowchart TD
    subgraph SENSING["Physical & Chemical Data Acquisition"]
        CAM["OV2640 DVP Camera\n(160x120 RGB565 via DCMI DMA)"]
        TCS["TCS34725 Colorimeter\n(16-bit Clear, R, G, B via I2C)"]
        MQ["MQ135 Gas Sensor\n(Volatile VOCs / Ethylene / Ammonia)"]
        PH["Analog Glass pH Probe\n(Nernst Wash-Water Acidification)"]
        DHT["DHT22 Sensor\n(Temperature & %RH Normalization)"]
    end

    subgraph MCU["STM32F407VET6 Processing Engine (168 MHz)"]
        GATE["Spatial Texture & Color Gate\n(Foreign Object & Hand Rejection)"]
        M1["Model 1: MobileNetV2-0.35 INT8\n(CMSIS-NN 80x80x3 Shared Backbone)"]
        M3["Model 3: Multi-Head Estimator\n(Quality Grade & Remaining Shelf Life)"]
        M2["Model 2: Sensor Fusion Classifier\n(Multinomial Logistic Regression Engine)"]
    end

    subgraph OUTPUT["Decision & User Interface"]
        OLED["SSD1306 0.96 inch I2C OLED\n(8-Row Dynamic UI / Telemetry Pages)"]
        ALERT["Auditory Buzzer & Safety Badges\n('>> SAFE TO EAT <<' / '>> WASH REQUIRED <<')"]
    end

    CAM --> GATE
    GATE -->|Passed Fruit Sample| M1
    TCS --> GATE
    TCS --> M2
    MQ --> M2
    PH --> M2
    DHT --> M2
```

```
    M1 --> M3
    M1 --> OLED
    M3 --> OLED
    M2 --> OLED
    M2 --> ALERT
```

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2. Hardware Architecture & Embedded Memory Allocation

The embedded firmware runs entirely bare-metal without dynamic heap allocation (`malloc` is strictly prohibited in runtime loops), ensuring deterministic execution and preventing fragmentation.

2.1 MCU Specifications & Memory Footprint

The STM32F407VET6 features **512 KB Flash ROM**, **128 KB contiguous Main SRAM** (SRAM1: 112 KB + SRAM2: 16 KB), and **64 KB Core Coupled Memory (CCMRAM)**.

```
pie title Memory Allocation on STM32F407VET6
    "Firmware & INT8 Model Weights (Flash 435 KB)" : 435
    "Free Flash ROM (77 KB)" : 77
    "DMA Camera & Vision Arena (Main SRAM 108 KB)" : 108
    "Free Main SRAM (20 KB)" : 20
    "Persistent TFLM Tensor Tail (CCMRAM 53 KB)" : 53
    "Free CCMRAM (11 KB)" : 11
```

Memory Region	Physical Address Range	Hardware Capacity	Firmware Usage	Utilization	Architectural Function
Flash ROM	`0x08000000` – `0x0807FFFF`	512 KB	435.3 KB	**85.0%**	Code `.text`, Vector Table, Read-only Lookup Tables, and Quantized INT8 Weights
Main SRAM (SRAM1+2)	`0x20000000` – `0x2001FFFF`	128 KB	108.2 KB	**82.6%**	DCMI DMA frame buffer, peripheral buffers, C runtime stack (8 KB), and head activation arena
CCMRAM	`0x10000000` – `0x1000FFFF`	64 KB	53.2 KB	**81.3%**	TFLite-Micro persistent runtime tensor descriptors and CMSIS-NN requantization tables

> [!IMPORTANT] **CCMRAM Bus Isolation:** CCMRAM is coupled directly to the Cortex-M4 D-bus and **cannot be accessed by the DMA2 controller**. Therefore, camera DMA transfers are strictly assigned to Main SRAM (`0x20000000`), while persistent neural network weights and lookup tables reside safely in Flash and CCMRAM.

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3. Two-Tier TinyML Machine Learning Pipeline

3.1 Model 1: Deep Computer Vision (INT8 Quantized MobileNetV2)

- **Input Pipeline:** Raw 160×120 RGB565 frames captured by the OV2640 camera are converted on-the-fly to $80 \times 80 \times 3$ RGB888 format using aspect-preserving bilinear interpolation and zero-padding.
- **Normalization:** In-model scaling: $x_{\text{norm}} = \frac{x}{127.5} - 1.0$ (eliminates floating-point overhead on the input buffer).
- **Backbone:** Truncated MobileNetV2 with width multiplier $\alpha = 0.35$, cut after layer `block_15_add` to eliminate memory-heavy expansion layers.
- **Quantization:** Fully integerized (INT8 weights and activations) executed using ARM CMSIS-NN SIMD instructions, achieving bit-identical numerical parity with desktop LiteRT execution.

3.2 Model 2: Multi-Sensor Chemical Fusion (Risk Scoring)

Model 2 processes physical and chemical sensor features into an estimated pesticide risk classification (`LOW`, `MEDIUM`, or `HIGH`). - **Feature Vector ($\mathbf{x} \in \mathbb{R}^{12}$):** 1. Optical raw channels: $\text{TCS}_R, \text{TCS}_G, \text{TCS}_B, \text{TCS}_C$ 2. Gas & Electrochemical: $\ln(\text{MQ135}), \text{pH}$ 3. Ambient normalization: $\text{Temperature}\ (^\circ\text{C}), \text{Relative Humidity}\ (\%RH)$ 4. Invariant optical

ratios: $\frac{R}{C}, \frac{G}{C}, \frac{B}{C}$ - **Inference Engine:** Multinomial Logistic Regression calibrated using Nernstian temperature offsets and z-score standard scaling: $$\mathbf{z} = \frac{\mathbf{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}, \quad \mathbf{y} = \text{softmax}(\mathbf{W}\mathbf{z} + \mathbf{b})$$ Because $\mathbf{W} \in \mathbb{R}^{3 \times 12}$, inference requires only 36 multiply-accumulate operations, completing in **under 50 microseconds**.

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4. Foreign Object Rejection & Biological Safeguards

To prevent false classifications when non-fruit objects (such as leather wallets, mobile phones, printed paper, or hands) are placed on the inspection tray, the firmware executes a real-time Laplacian texture filter and optical band check before invoking neural network inference:

$$\text{Image Detail} = \sum_{x,y} \left(|I(x,y) - I(x+1,y)| + |I(x,y) - I(x,y+1)| \right)$$

```
flowchart TD
    FRAME["Incoming Camera Frame (OV2640)"] --> COND1{"Confidence < 0.35 OR\nImage Detail > 1400.0?"}
    COND1 -->|Yes: Synthetic/Texture Violation| REJ["Set Flag: FV_IC_REJ_SKIN_OBJECT\nFruit ID = -1 (Rejected)\nDisplay: [ INVALID SAMPLE ] Non
    COND1 -->|No: Botanical Smoothness Valid| OPT{"TCS Blue Plastic Violation?\n(B > 3.5*R AND B > 1.8*G)"}
    OPT -->|Yes: Synthetic Pigment| REJ
    OPT -->|No| SPEC{"Species Disambiguation"}
    SPEC -->|Apple predicted BUT G/R > 1.8| GUAVA["Reclassify: Guava (Chlorophyll Peak)"]
    SPEC -->|Dark produce AND B > 1.3*R| BRINJAL["Reclassify: Brinjal (Anthocyanin Peak)"]
    SPEC -->|Standard profile| NATIVE["Retain Native Vision Output"]
```

Measured Surface Texture Index vs. Fruit Skin

- **Ripe Tomato:** `img_detail = 201.7`
- **Banana:** `img_detail = 320.5 – 464.7`
- **Guava:** `img_detail = 432.9`
- **Apple:** `img_detail = 567.1`
- **Brinjal (Calyx):** `img_detail = 883.3 – 1190.7`
- **Leather Wallet / Artificial Fabric:** **`img_detail = 2472.4`** $\rightarrow$ **Cleanly triggers rejection gate (> 1400.0).**

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5. Official Data Tables (Report Chapters 7 & 11)

Table 7.3: Summary of the Collected Dataset

Target Fruit	Source Repository	Usable Samples	Shelf-Life Testing Window	Quality Label Scheme
Apple	FruitNet (Mendeley Data)	2,275	8–10 days	Binary Fresh / Spoiled
Banana	FruitNet & AgriFreshNET	2,674	13 days	3-Stage: Fresh (1–4d), Semi (4–7d), Rotten (7–13d)
Guava	FruitNet (Mendeley Data)	2,281	7 days	Class A / Class B / Defect
Tomato	AgriFreshNET (Mendeley Data)	838	35 days	3-Stage: Fresh (1–10d), Semi (10–24d), Rotten (24–35d)
Brinjal	AgriFreshNET (Mendeley Data)	389	15 days	3-Stage: Fresh (1–4d), Semi (4–8d), Rotten (8–15d)
Total	*Crated Multi-Source Benchmark*	**8,457**	—	**6,146 Train / 1,299 Val / 1,012 Test**

Table 7.4: Average Sensor Readings of Each Fruit (30-Reading Averages)

Fruit	Temp (°C)	Humidity (%)	pH	MQ135 (ADC)	TCS R	TCS G	TCS B	Detected Colour
Apple	28.5	54.2	6.42	685	625	430	410	Red
Banana	28.8	52.6	6.15	740	785	720	390	Yellow
Guava	28.2	55.1	5.85	620	420	780	380	Green

Fruit	Temp (°C)	Humidity (%)	pH	MQ135 (ADC)	TCS R	TCS G	TCS B	Detected Colour
Tomato	28.1	56.4	6.30	580	840	390	360	Red
Brinjal	28.6	53.8	6.65	710	380	320	690	Violet

Table 7.5: TinyML Models and Performance Benchmark

Model ID	Input Representation	Target Output Classes	Test Accuracy	Macro-F1 Score
Model 1: Vision	Camera frame (160×120), cropped & resized to $80 \times 80 \times 3$	5 Fruit Species + Binary Freshness Condition	**79.6%** *(Fruit)* **68.1%** *(Condition)*	**82.2%** *(Fruit)* **67.8%** *(Condition)*
Model 2: Fusion	12 Physical & Chemical Sensor Ratios	Chemical Residue Risk (`LOW`, `MED`, `HIGH`)	**78.6%**	**77.2%**

Table 7.6: Embedded Microcontroller Latency and Memory Utilization

Model	On-Chip Footprint (Flash ROM)	Peak Working RAM (Main SRAM)	Persistent Arena (CCMRAM)	Inference Execution Latency
Model 1 *(MobileNetV2-0.35 INT8)*	326.0 KB	96.0 KB	53.2 KB	**120 ms** (@ 168 MHz)
Model 2 *(Sensor Fusion Logistic Reg.)*	0.25 KB	0.06 KB	0.0 KB	**< 1 ms** (~0.05 ms)

Table 7.7: OLED Display Final Output Page Verification

Tested Produce	Classification	Freshness & Quality	Shelf Life	Safety Level	Safety Badge & Wash Action
Apple	Apple	Fresh (81%, Good)	8 Days	Moderate Risk	`>> WASH REQUIRED <<` Warm Soda Dip (Removes wax)
Banana	Banana	Fresh (82%, Good Ripe)	4 Days	Moderate Risk	`>> WASH REQUIRED <<` Peel Wash Advisory
Guava	Guava	Fresh (80%, Good)	6 Days	Moderate Risk	`>> WASH REQUIRED <<` 2% Salt Solution Dip
Tomato	Tomato	Fresh (81%, Good Organic)	6 Days	Low Risk	`>> SAFE TO EAT <<` Farm Fresh Organic (94%)
Brinjal	Brinjal	Fresh (80%, Good)	4 Days	Moderate Risk	`>> WASH REQUIRED <<` Salt Brine Soak

Table 11.3: Recorded Multi-Sensor Produce Dataset (Appendix C Excerpt)

sample_id	fruit	source	treatment	temp_C	hum_pct	pH	ph_volt	mq135	gas_ratio	R	G	B	colour_name	dark_
SMP-001	Apple	Market	Wax coated (Commercial)	30.8	46.3	6.15	2.110	776	1.49	672	429	380	Red	0
SMP-003	Apple	Market	Untreated (Market)	31.1	50.2	6.21	2.111	785	1.51	635	439	402	Red	0
SMP-005	Apple	Organic	None (Certified Organic)	29.5	48.1	6.52	2.062	668	1.28	664	422	415	Red	0

sample_id	fruit	source	treatment	temp_C	hum_pct	pH	ph_volt	mq135	gas_ratio	R	G	B	colour_name	dark_
SMP-007	Apple	Pesticide-applied	Chlorpyrifos 0.2% spray	30.4	49.7	5.38	2.138	985	1.89	658	435	382	Red	0
SMP-009	Apple	Market	Stored 12 days (Decay)	31.2	53.8	4.98	2.155	1110	2.13	648	415	376	Red	2
SMP-011	Banana	Market	Ethylene ripened	30.5	48.9	6.14	2.098	895	1.72	775	742	385	Yellow	0
SMP-015	Banana	Organic	Naturally tree-ripened	31.6	49.5	6.45	2.068	705	1.36	796	748	370	Yellow	0
SMP-017	Banana	Pesticide-applied	Carbofuran surface dip	29.8	52.1	5.21	2.145	1055	2.03	780	735	392	Yellow	0
SMP-019	Banana	Market	Stored 8 days (Senescent)	31.5	48.2	5.08	2.168	1210	2.33	742	710	360	Yellow	3
SMP-021	Guava	Market	Standard harvest	29.4	53.8	5.85	2.112	740	1.42	422	782	380	Green	0
SMP-025	Guava	Organic	Orchard organic	28.7	52.8	6.15	2.091	610	1.17	412	798	365	Green	0
SMP-027	Guava	Pesticide-applied	Malathion 0.1% spray	30.6	50.5	5.02	2.165	925	1.78	405	760	372	Green	0
SMP-031	Tomato	Organic	None (Farm Fresh Organic)	28.3	55.0	6.58	2.068	560	1.08	865	395	342	Red	0
SMP-033	Tomato	Market	Commercial harvest	29.3	52.7	6.25	2.085	715	1.38	842	402	355	Red	0
SMP-037	Tomato	Pesticide-applied	Dimethoate 0.2% spray	31.2	48.6	5.32	2.145	948	1.82	835	382	365	Red	0
SMP-041	Brinjal	Market	Commercial market	30.4	51.2	6.65	2.065	790	1.52	372	325	698	Violet	0
SMP-045	Brinjal	Organic	Local polyhouse organic	28.6	54.2	6.88	2.050	635	1.22	390	338	712	Violet	0
SMP-047	Brinjal	Pesticide-applied	Cypermethrin 0.15% spray	30.8	49.9	5.62	2.128	1025	1.97	362	312	672	Violet	0
SMP-049	Brinjal	Market	Stored 10 days (Wrinkled rot)	32.3	47.8	5.22	2.152	1155	2.22	350	302	638	Violet	5

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6. Comprehensive Viva Voce Defense & Examination Q&A

Q1: How can an embedded microcontroller perform deep learning without a dedicated GPU or NPU?

Answer: The STM32F407VET6 utilizes **TensorFlow Lite for Microcontrollers (TFLM)** with **ARM CMSIS-NN** (Cortex Microcontroller Software Interface Standard - Neural Network) optimized kernel libraries. CMSIS-NN exploits the Cortex-M4's SIMD (Single Instruction, Multiple Data) instructions (such as `SMLAD` - Signed Multiply Accumulate Dual) to perform four 8-bit integer multiply-accumulate operations in just two clock cycles. By fully quantizing weights and activations from 32-bit floating point down to symmetric signed 8-bit integers (`int8`), memory bandwidth is reduced by 75% and mathematical throughput increases by nearly $4\times$, enabling complete inference in 120 ms at 168 MHz.

Q2: What is the scientific principle behind using pH and MQ135 to detect pesticide residue?

Answer: Common agricultural insecticides belong predominantly to the **organophosphate** (e.g., Chlorpyrifos, Dimethoate) and **carbamate** (e.g., Carbofuran) chemical classes. 1. **pH Acidification:** During aqueous washing, organophosphates undergo hydrolysis, liberating acidic phosphoric and thiophosphoric acid derivatives that lower wash-water pH from neutral (~7.0) down into the 5.2–5.6 range. The potentiometric glass electrode detects this H^+ ion gradient following the Nernst equation: $$E = E_0 - \frac{2.303 R T}{F} \cdot \text{pH}$$ 2. **Volatile Gas Detection (MQ135):** Organophosphate and carbamate solvents release volatile organic compounds (VOCs) and sulfur/nitrogen heteroatoms. When these contact the tin dioxide (SnO_2) semiconductor surface at 300°C, oxygen ions release trapped electrons, causing an immediate drop in sensor resistance (R_s) and a spike in ADC voltage.

Q3: Why does the system classify the Tomato as "Low Risk (Organic Safe)" while Banana and Apple are classified as "Moderate Risk"?

Answer: In commercial post-harvest supply chains, apples receive carnauba/shellac wax coatings containing fungicide residues (e.g., Imazalil, Thiabendazole), while bananas are commercially treated with ethylene and post-harvest antifungal washes, exhibiting elevated baseline gas readings ($\approx 740\text{ ADC counts}$) and requiring a mandatory wash advisory. Conversely, certified organic tomatoes originate from chemical-free soil, yielding clean baseline VOC levels ($\approx 580\text{ counts}$) and neutral wash-water pH ($6.30 - 6.58$), safely qualifying for direct raw consumption (`Safe To Eat`).

Q4: How does the system reject foreign non-fruit objects (e.g., a wallet or hand) from being classified as a fruit?

Answer: Closed-set softmax neural networks inherently produce a high-probability classification even when fed completely out-of-distribution inputs. To overcome this vulnerability, the system enforces a multi-layer spatial and optical consistency gate before invoking the classification head: 1. **Spatial Laplacian Gradient:** Botanical produce skins are smooth, yielding low high-frequency spatial gradients ($\text{img_detail} < 1200$). Micro-textured artificial surfaces like leather grain, zippers, paper fibers, or skin prints produce intense high-frequency variance ($\text{img_detail} = 2472.4$). Thresholding at $\text{image_detail} > 1400.0$ immediately traps non-fruit objects. 2. **Synthetic Optical Signature:** Blue pigments are practically non-existent in terrestrial fruit skin. If TCS34725 readings detect unnatural blue dominance ($B > 3.5 \times R$ and $B > 1.8 \times G$), the sample is flagged as artificial plastic/ink and rejected with `>> SCAN REJECTED <<`.

Q5: Why can't CCMRAM be used for camera DMA buffers?

Answer: The STM32F407 interconnect architecture links the Core Coupled Memory (CCMRAM) strictly to the Cortex-M4 D-bus. The multi-master AHB bus matrix does **not** grant the general DMA2 controller direct access to CCMRAM. Attempting to target a camera DCMI DMA transfer into CCMRAM results in an immediate DMA transfer error (`TEIF` interrupt). Hence, Main SRAM (`0x20000000`) is dedicated to DMA ping-pong buffers, while CCMRAM is safely populated with static neural network operator tables and scratch structures.

Verification & Deliverable Sign-Off

- **Firmware Binary:** `reports/firmware/gcc/fv_app_final_presentation.bin` (Flashed via ST-Link at `0x08000000`)
- **Regression Suite:** 180 tests passing (`pytest tests/`)
- **Repository Branch:** `main` (Commit `9484e8d`)